

APPENDIX TO THE REQUIREMENT GATHERING DOCUMENT (RGD v2.0)

GAP-14

Appendix-N

Deployment & Infrastructure Architecture

School ERP

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Classification: Confidential — Client & Project Team Only

1. Purpose

This document is Appendix-N to the Requirement Gathering Document (RGD v2.0) for the School ERP System, addressing Gap-14: Deployment & Infrastructure Architecture. It defines the logical infrastructure — environments, topology, layers, and operational strategy — needed to host and operate the system whose business requirements, architecture, internal design, external interfaces, deployment principles, and test strategy were established across the preceding thirteen documents (RGD v2.0 and Appendix-A through Appendix-M).

This is a business and logical infrastructure document, not an implementation artifact. It does not name a specific cloud provider, and it does not produce source code, SQL, API definitions, JSON/XML, Terraform, Kubernetes manifests, Docker Compose files, shell commands, or configuration files of any kind. It describes infrastructure shape and operational principles in a technology-neutral, vendor-neutral form.

Per the governing rules for this task, this document does not regenerate RGD v2.0 or any prior appendix. Every reference below to a Non-Functional Requirement, architectural layer, internal component, API group, deployment principle, or test-strategy element is reused by ID from Appendix-F, Appendix-H, Appendix-I, Appendix-J, Appendix-K, Appendix-L, and Appendix-M, which remain authoritative.

1.1 Relationship to Appendix-L

Appendix-L (Deployment Architecture & Infrastructure Design) already established the system's logical deployment topology, environment model, and operational strategy at a foundational level. This Appendix-N does not duplicate or contradict Appendix-L — it restates and extends that same logical infrastructure under the specific 30-section structure and environment set (including Training, distinct from UAT and Staging) requested for this document, consolidating it alongside Appendix-M's Test Strategy for a single, traceable infrastructure-and-quality reference.

1.2 Document Structure

§	Content
1-4	Purpose, scope, references, and deployment objectives
5-10	Environment strategy and the core infrastructure layers (application, database, storage)
11-14	Network, load balancing, availability, and scalability
15-17	Disaster recovery, backup, and business continuity
18-20	Identity/access, security zones, and encryption (logical)
21-23	Logging, monitoring, alerting, and capacity planning
24-26	Release flow, environment promotion, and operational ownership
27-30	Validation, risks/assumptions, traceability, and final review

2. Scope

2.1 In Scope

- Logical infrastructure architecture across six environments: Development, Testing, UAT, Training, Staging, Production.
- Application, Database, and File Storage layer responsibilities; network segmentation and security zones.
- High Availability, Scalability, Disaster Recovery, Backup, and Business Continuity strategy.
- Identity & Access infrastructure, encryption strategy (logical only), logging, monitoring, and alerting architecture.
- Release deployment flow, environment promotion process, and operational responsibilities.

2.2 Out of Scope

- Source code, SQL, API definitions, or JSON/XML payloads of any kind.
- Terraform, Kubernetes manifests, Docker Compose files, shell commands, or configuration files.
- Any cloud-specific implementation detail or vendor selection — every such decision is marked Client / Infrastructure Team Decision Required throughout.
- Low Level Design, database schema, and test-case-level detail — remain the domain of Appendix-J, Appendix-H, and Appendix-M respectively, unmodified here.

3. References (RGD + Appendix A–M)

Every document in this set is reused by reference only. None is regenerated, rewritten, or reinterpreted in producing this Appendix-N.

Document	Relevance to This Appendix
RGD v2.0	Business objectives, scope, and Business Assumptions (Section 22, cloud-hosting assumption)
Appendix-A (BPMN)	Process context underpinning Application Layer responsibilities (Section 8)
Appendix-B (RTM)	Requirement traceability underpinning Section 29
Appendix-C v1.1 (Business Rules)	Regulatory and Critical-priority rules referenced in Security Zones (Section 19)
Appendix-D (RDM)	Module dependency chain referenced in Application Layer (Section 8)
Appendix-E (Functional Requirements)	Functional scope underpinning Capacity Planning (Section 23)
Appendix-F (NFR Catalogue)	Primary source for Sections 13-23 (Availability, Scalability, DR, Backup, Security, Monitoring, Performance)
Appendix-G (Data Dictionary)	Entity/module structure underpinning Database Layer (Section 9)
Appendix-H (ERD & Schema Design)	Logical data model, partitioning, and normalization underpinning Section 9
Appendix-I (HLD)	System architecture baseline for Sections 6-8, 18-19
Appendix-J (LLD)	Internal component design underpinning Application Layer (Section 8)
Appendix-K (API Specification)	External interface shape referenced in Load Balancing (Section 12)
Appendix-L (Deployment Architecture)	Direct architectural baseline for this entire document (Section 1.1)
Appendix-M (Test Strategy & QA)	Test environment mapping cross-referenced in Environment Strategy (Section 5)

4. Deployment Objectives

Reused directly from Appendix-F's Non-Functional Requirement targets, consistent with Appendix-L Section 3 — this document does not set new targets, only the infrastructure needed to meet them.

Objective	Target (per Appendix-F)
Availability	99.5% uptime during academic operating hours (NFR-AVAIL-001)
Reliability	<0.1% critical-transaction error rate (NFR-REL-001)
Recovery Time Objective (RTO)	≤4 hours (NFR-BKP-003, Client / Infrastructure Team Decision Required)
Recovery Point Objective (RPO)	≤1 hour (NFR-BKP-004, Client / Infrastructure Team Decision Required)
Concurrent User Support	2,000 (NFR-SCAL-001, Client / Infrastructure Team Decision Required)
Page Load / API Response	≤3 seconds / ≤500ms respectively (NFR-PERF-001/011)
Test-Verified Quality Bar	Per Appendix-M Section 22 Acceptance Criteria before Production promotion

5. Environment Strategy

Six environments, extending Appendix-L Section 4's four-environment model with two additional environments explicitly requested for this document: UAT as a distinct client-validation environment separate from Staging, and Training as a dedicated environment for staff onboarding rehearsal.

Environment	Purpose	Data
Development	Active feature development and Unit Testing (Appendix-M Section 8)	Synthetic/anonymized test data only
Testing	Integration and System Testing against Appendix-E acceptance criteria (Appendix-M Section 8)	Synthetic test data
UAT	Client business-scenario validation against RGD v2.0 and Appendix-E (Appendix-M Section 8)	Anonymized production-like or client-provided sample data (Client / Infrastructure Team Decision Required)
Training	Staff onboarding and hands-on rehearsal ahead of go-live, isolated from UAT sign-off activity	Synthetic/demonstration data only, reset between training sessions
Staging	Final pre-production rehearsal — Non-Functional Testing (Appendix-M Section 10), Disaster Recovery drills (Section 15)	Production-representative anonymized data (Client / Infrastructure Team Decision Required)
Production	Live operation	Real student/staff/financial data, subject to full Security Zones (Section 19)

5.1 Why Training Is Separated from UAT

UAT is a sign-off-bearing activity (Project Sponsor acceptance, per RGD v2.0's Approval Matrix and Appendix-M Section 22), while Training is a repeated, low-stakes rehearsal activity for Admin Staff, Teachers, and other end users (Appendix-I Section 8 User Roles) to build familiarity before go-live. Combining them into a single environment would risk either disrupting UAT sign-off evidence with training activity, or constraining Training scheduling around UAT's more formal cadence — separating them removes both risks.

5.2 Environment Promotion Principle

Consistent with Appendix-L Section 4.1: code and configuration changes flow Development → Testing → UAT → Staging → Production in sequence. Training runs in parallel to UAT/Staging rather than in this promotion chain, since it does not gate any release decision — it is a people-readiness activity, not a system-quality gate.

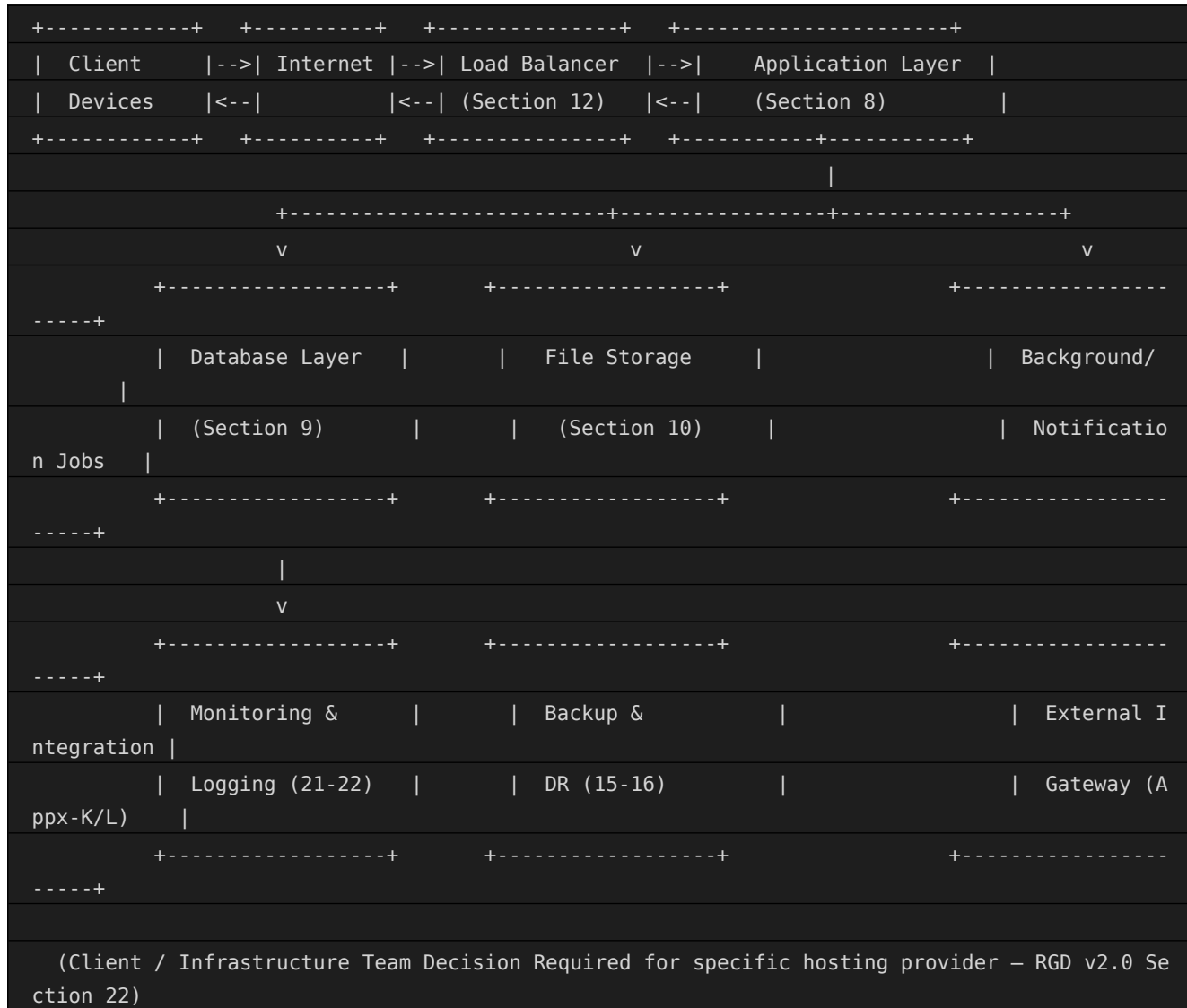
5.3 Infrastructure Scale by Environment

Environment	Application Layer	Database Layer
Development	Single instance	Single instance, no replica

Environment	Application Layer	Database Layer
Testing	Single instance	Single instance, no replica
UAT	Multiple instances, reduced count	Primary + replica, reduced capacity
Training	Single or reduced-count instance	Single instance, reset-friendly configuration
Staging	Multiple instances, production-representative	Primary + replica, production-representative
Production	Multiple instances, full count (Section 14)	Primary + replica, partitioned (Section 9)

6. Logical Infrastructure Architecture

Consolidates the infrastructure components already defined in Appendix-L Section 6, presented here as the first required ASCII diagram for this document.



6.1 Infrastructure Component Inventory

Component	Role
Client Devices	Desktop, tablet, and mobile browsers (Appendix-F NFR-BRC-001, NFR-MOB-001)
Load Balancer	Distributes requests across Application Layer instances (Section 12)
Application Layer	Hosts Presentation and Business Logic (Appendix-I Section 7; Appendix-J Sections 5-6)
Database Layer	Hosts the 39 entities defined in Appendix-H (Section 9)

Component	Role
File Storage	Hosts Document entity artifacts (Appendix-G/H ENT-SYS-006; Section 10)
Background/Notification Jobs	Scheduled and event-driven processes (Appendix-J Section 17; Appendix-L Section 6.9-6.10)
Monitoring & Logging	Real-time visibility and centralized log aggregation (Sections 21-22)
Backup & DR Infrastructure	Automated backup and failover capability (Sections 15-16)
External Integration Gateway	Boundary to the seven external integrations (Appendix-I Section 11; Appendix-K Section 17)

6.2 Component Interaction Summary

The 9 components above are not independent silos — each interacts with its neighbors exactly as shown in the diagram, and each maps onto exactly one Security Zone (Section 19). No component has a direct dependency on a component more than one tier away (e.g., the Load Balancer never calls the Database Layer directly), preserving the same layered-isolation principle already established at the application level in Appendix-I Section 7 and reused throughout Appendix-L.

7. Deployment Topology

Reuses Appendix-L Section 5's tier structure, restated here as the system's overall deployment shape independent of any specific environment (Section 5 covers per-environment scaling detail).

- Every tier in Section 6.1 is deployed independently and scales independently — the Application Layer scales horizontally (Section 14) without requiring a corresponding change to the Database Layer's topology, and vice versa.
- No tier is bypassed — a request from a Client Device always traverses Load Balancer → Application Layer → (Database Layer / File Storage / Background Jobs) in that order, never directly reaching a downstream tier.
- The External Integration Gateway is topologically parallel to, not part of, the core request-serving path — an external vendor outage (Appendix-I Section 11.2) does not block core request-serving capability for functionality that does not depend on that specific integration.

7.1 Topology Consistency Across Environments

The same topology shape applies to every environment in Section 5, differing only in scale and redundancy (Section 5 table "Data" column notwithstanding) — Development runs a minimal, single-instance version of the same topology shown in Section 6, not a structurally different architecture, consistent with the Environment Parity principle already established in Appendix-L Section 4.2 and reused in Appendix-M Section 11.1.

8. Application Layer

Reuses Appendix-I Section 7 (Layered Architecture) and Appendix-J Sections 5-6 (Component Breakdown, Controller-Service-Repository Interaction) at the infrastructure level — this section describes where and how that design is hosted, not what it contains.

Responsibility	Description
Presentation Serving	Renders the Admin/Staff portal, Parent/Student portal, and mobile-responsive views (Appendix-I Section 7)
Business Logic Execution	Hosts Controller-Service-Repository components for all 11 modules (Appendix-J Sections 4-6)
API Group Serving	Serves the conceptual API Groups catalogued in Appendix-K Section 18
Authentication & Authorization Enforcement	Token validation and Role Permission checks at the Controller boundary (Appendix-K Sections 4-5; Section 18 this document)
Statelessness	No server-side session state beyond the authenticated identity token (Appendix-I Section 18.1), enabling horizontal scaling (Section 14)
Background Job Hosting	Executes the 8 scheduled/triggered processes catalogued in Appendix-J Section 17, logically separate from request-serving capacity

9. Database Layer

Reuses Appendix-H's logical data model (39 entities, 45 relationships) and Appendix-L Section 10 deployment principles.

Responsibility	Description
Primary Data Persistence	Read/write instance hosting all 39 entities (Appendix-H)
Read Scaling	At least one read replica, offloading reporting-heavy queries (Appendix-K Section 18, Reports module)
Partitioning	AttendanceRecord, AuditLog, NotificationLog, Invoice/Payment — partitioned per Appendix-H Section 17.8
Referential Integrity Enforcement	Foreign Key constraints per Appendix-H Section 11, with the polymorphic-association mitigation noted in Appendix-J Section 22
Connection Management	Application Layer components pool database connections rather than opening one per request (Client / Infrastructure Team Decision Required for pool sizing)
Growth Accommodation	Sized against the Data Volume Estimates in Appendix-H Section 17.7 (10M+ AttendanceRecord/year, 15M+ AuditLog/year estimated)

9.1 Database Deployment by Environment

Development and Testing run a single, non-replicated instance, sufficient for functional validation. UAT and Training similarly run reduced-scale instances, since neither requires full replica topology. Staging runs the full primary+replica topology at reduced capacity specifically to rehearse failover (Section 15) realistically before Production go-live.

10. File Storage Strategy

Reuses Appendix-I Section 12 and Appendix-L Section 11, describing the Document entity's (Appendix-G/H ENT-SYS-006) storage infrastructure.

Storage Class	Content	Retention Approach
Active/Hot Storage	Documents linked to Active entities (current-session Invoices, active Student profiles)	Retained per the owning entity's active lifecycle (Appendix-G Field 19)
Archival/Cool Storage	Documents linked to Archived entities (closed-session Invoices, exited Students)	Per Appendix-H Section 15 Archival Policy, aligned with the owning entity's schedule
Generated Documents	System-generated artifacts (report cards, ID cards, payslips, receipts)	Regenerable from source data, but retained as issued records for audit (Appendix-J Section 16)

Object/file storage is logically separate from the Database Layer — Document.file_path is a reference, not an embedded binary (Appendix-I Section 12). Retrieval is always mediated through an authorized query operation (Appendix-I Section 12.3), never a raw, unauthenticated storage path. Specific storage provider and tiering configuration remain Client / Infrastructure Team Decision Required.

11. Network Segmentation

Reuses Appendix-L Section 7 (Network Architecture), restated here under this document's own section numbering as a network path table rather than a repeated diagram (the equivalent flow is shown pictorially in the Logical Infrastructure Architecture diagram, Section 6).

Hop	From	To	Zone Transition (Section 19)
1	Client Devices	Public Network / Internet	N/A — external to the system
2	Internet	Load Balancer	Entering Public Zone
3	Load Balancer	Application Layer	Public Zone → Application Zone
4	Application Layer	Database Layer / File Storage	Application Zone → Restricted Zone
5	Application Layer	External Integration Gateway	Application Zone → Integration Zone (outbound-only to Payment/SMS-Email/GPS/UIDAI/Tally/SSO, all vendors Client / Infrastructure Team Decision Required)

- No tier below the Load Balancer is directly reachable from the public Internet — every request is mediated through the tier immediately above it, consistent with Appendix-I Section 7.3's Presentation-to-Application boundary principle.
- Segmentation follows the Security Zone model (Section 19), bounding the blast radius of a compromise at any single tier.

12. Load Balancing Strategy

Reuses Appendix-L Section 6.3 and Appendix-I Section 17.3, elaborated here as its own section per this document's structure.

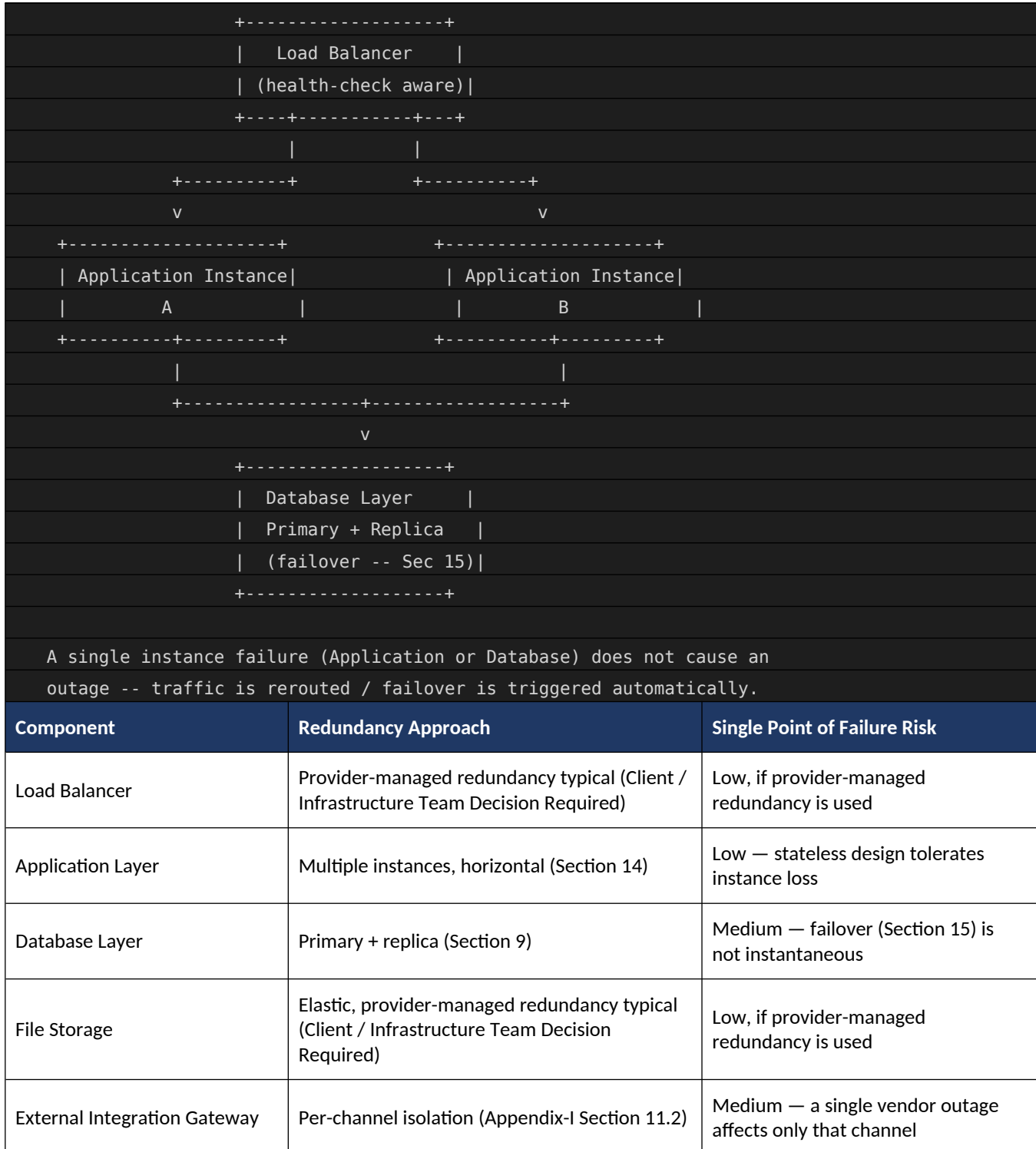
- Distributes incoming requests across Application Layer instances, enabling the horizontal scaling described in Section 14.
- Performs health checks against Application Layer instances, routing traffic away from an unhealthy instance automatically — this is the primary mechanism supporting the High Availability strategy in Section 13.
- Serves as the typical TLS/encryption termination point (Section 20), though exact placement of termination relative to internal-network encryption remains Client / Infrastructure Team Decision Required.
- Load balancing applies uniformly across the API Groups already catalogued in Appendix-K Section 18 — no API Group receives dedicated, separate load-balancing treatment.

12.1 Load Balancing by Environment

Development and Testing typically run a single Application Layer instance, making load balancing largely moot in those environments. UAT, Training, Staging, and Production run multiple instances behind a Load Balancer, with instance count scaled per environment purpose (Section 5).

13. High Availability Strategy

Reuses Appendix-F's Availability NFR series (NFR-AVAIL-001/002) and Appendix-L Section 14, presented here with the required High Availability Architecture diagram.



The 99.5% uptime target (NFR-AVAIL-001) is measured on a rolling 90-day window; planned maintenance is limited to ≤ 4 hours/month with 72-hour advance notice (NFR-AVAIL-002), scheduled to avoid academic operating hours.

14. Scalability Strategy

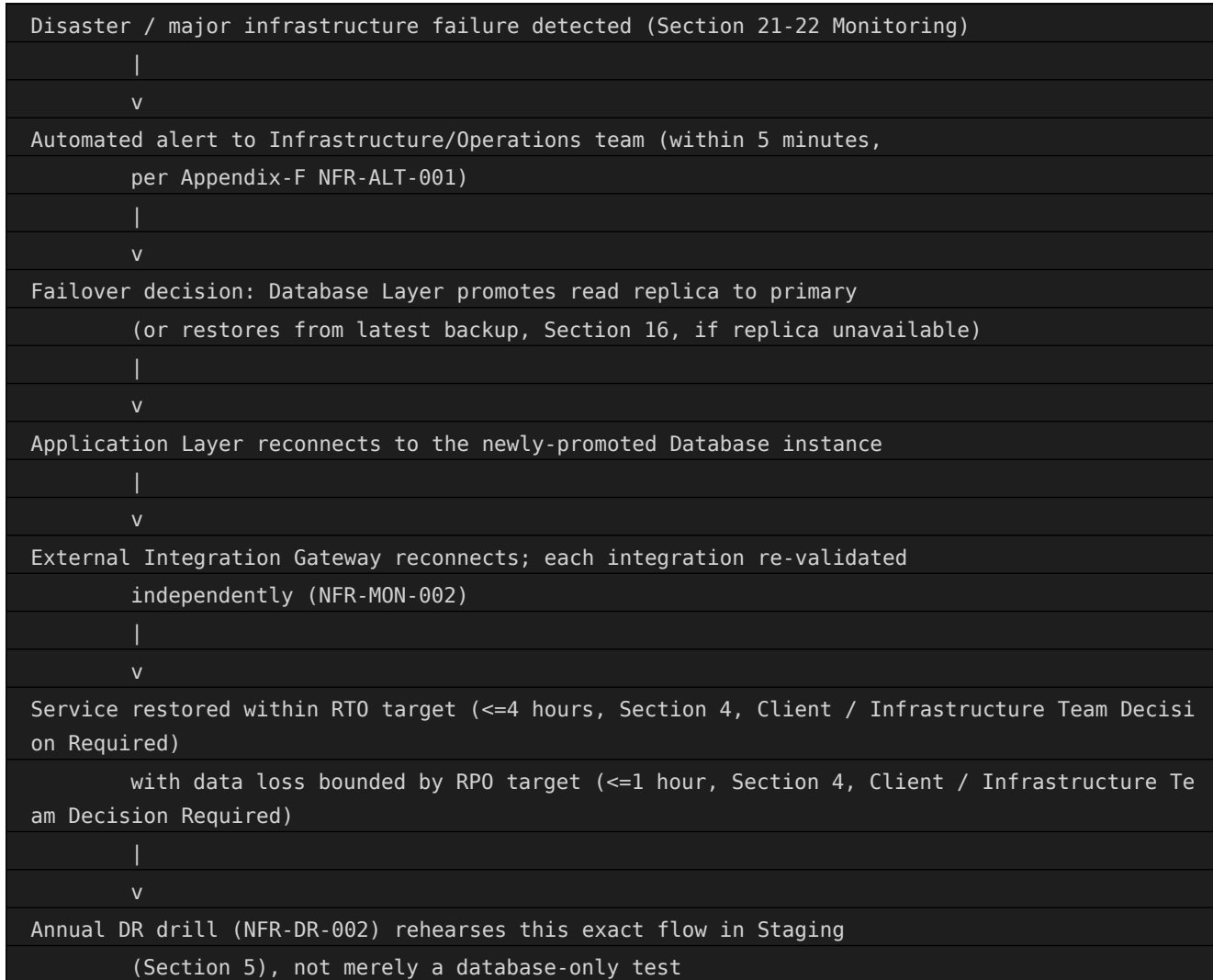
Reuses Appendix-F's Scalability NFR series (NFR-SCAL-001 through 008) and Appendix-L Section 15.

Tier	Scaling Approach
Application Layer	Horizontal — additional instances behind the Load Balancer (Section 12) as concurrent-user load increases
Database Layer (Primary)	Vertical scaling, with read-replica offload (Section 9)
Database Layer (High-Volume Tables)	Partition-level scaling (Section 9; Appendix-H Section 17.8)
File Storage	Elastic by nature of object storage (Section 10)

Capacity planning targets a 5-year data growth horizon without architectural redesign (Appendix-H Section 17.6), requiring the partition and archival strategy (Sections 9-10) to be operational from go-live, not retrofitted once volume becomes a visible problem. Scaling reviews are timed to the AcademicSession lifecycle boundary (Appendix-H), since enrollment and transaction volume step-change at year start rather than growing smoothly.

15. Disaster Recovery Strategy

Reuses Appendix-F's Disaster Recovery NFR series (NFR-DR-001/002) and Appendix-L Section 13, presented here together with Backup Strategy (Section 16) as the required combined Backup & Disaster Recovery Flow diagram.



15.1 Recovery Priority by Component

Component	Recovery Priority	Rationale
Database Layer	Critical — first	Every module depends on it; no other tier is useful without it
Application Layer	Critical — second	Restores request-serving capability once data is available
Load Balancer / Client Access	High — third	Restores client access once backend is functional
File Storage	Medium	Required for document-dependent operations, but core

Component	Recovery Priority	Rationale
		flows can proceed briefly without it
External Integration Gateway	Lower	Delayed notifications/payments are recoverable via retry (Appendix-J Section 17)

Manual-workaround procedures for Admission, Attendance, and Fee Collection (Appendix-F NFR-BC-001; Section 17 this document) remain available during an extended outage, independent of this technical failover process.

16. Backup Strategy

Reuses Appendix-F's Backup NFR series (NFR-BKP-001 through 005) and Appendix-L Section 12.

Component	Backup Method	Frequency
Database Layer (primary)	Full snapshot + continuous transaction log	Daily full, continuous log (NFR-BKP-001)
File Storage (Document entities)	Object-level replication or scheduled snapshot	Daily, aligned with Database backup window
Configuration data	Included in Database Layer backup (standard entity, Appendix-G ENT-SYS-005)	Same as Database Layer
Application release artifacts	Versioned artifact retained per Section 24 (recoverable via redeployment)	Per release, not scheduled

16.1 Recovery Objectives (RTO/RPO)

Metric	Target	Source
Recovery Time Objective (RTO)	≤4 hours (Client / Infrastructure Team Decision Required to confirm)	Appendix-F NFR-BKP-003
Recovery Point Objective (RPO)	≤1 hour (Client / Infrastructure Team Decision Required to confirm)	Appendix-F NFR-BKP-004
Backup Retention	30 days rolling + monthly archive for 1 year (Client / Infrastructure Team Decision Required to confirm)	Appendix-F NFR-BKP-002
Backup Integrity Verification	Monthly automated restore test	Appendix-F NFR-BKP-005

17. Business Continuity Considerations

Reuses Appendix-F NFR-BC-001 and Appendix-L Section 19 (referenced there under Availability/DR), addressed here as its own section.

- Manual-workaround procedures exist for the three highest-criticality processes — Admission, Attendance, and Fee Collection — allowing the school to continue essential operations during an extended technical outage, reusing the Workaround field already documented per requirement in Appendix-D.
- Business Continuity is distinct from Disaster Recovery (Section 15): DR restores the technical system; Business Continuity ensures the school can operate even while DR is in progress.
- Staff awareness of manual-workaround procedures is supported by the Training environment (Section 5), which can rehearse these procedures alongside normal system training.
- Communication to parents/students during an extended outage is handled through whatever channel remains available (e.g., direct phone/email outside the system), since the Communication module (Appendix-J Section 4.9) itself may be affected by the same outage.

18. Identity & Access Infrastructure

Reuses Appendix-K Section 4-5 (Authentication, Authorization) and Appendix-L Section 9, describing the infrastructure hosting these flows.

- The identity/token-issuing component resides in the Application Layer (Administration module, Appendix-J Section 4.11) — a Service-layer responsibility, not a separate infrastructure tier.
- Token validation occurs at the Application Layer's Controller boundary (Appendix-J Section 6) on every request, per Appendix-K Section 4's Authentication Flow.
- Role Permission resolution (Appendix-E Field 38; Appendix-I Section 8.3) occurs immediately after authentication, before any Business Logic executes, per Appendix-K Section 5's Authorization Flow.
- Account lockout state (NFR-PWD-003) and session/token expiry (NFR-SESS-001) are stored in the Database or Cache Layer, with the specific choice Client / Infrastructure Team Decision Required.

18.1 Access Infrastructure by Environment

Environment	Authentication Posture
Development	Simplified/local authentication acceptable, given synthetic data only
Testing	Full authentication flow, against synthetic test accounts
UAT	Full authentication flow, client-facing test accounts under Project Sponsor oversight
Training	Simplified, reset-between-session accounts appropriate for repeated rehearsal use
Staging	Full authentication flow, production-representative, including lockout/expiry behavior
Production	Full authentication flow, full security posture (Section 19-20), no exceptions

19. Security Zones

Reuses Appendix-I Section 16.2 and Appendix-L Section 8, mapped onto the infrastructure components in Section 6.1.

Zone	Components Included	Access Posture
Public Zone	Load Balancer (public-facing endpoint only)	Internet-reachable; encryption-terminated (Section 20); no application logic or data resides here
Application Zone	Application Layer	Reachable only from the Load Balancer; hosts Presentation and Business Logic
Restricted Zone	Database Layer, File Storage	Reachable only from the Application Layer; never directly Internet-facing
Integration Zone	External Integration Gateway	Outbound-only to external vendors; inbound only for vendor callbacks (e.g., payment confirmation)
Management Zone	Monitoring, Logging, Backup infrastructure	Accessible only to Infrastructure/Operations roles (Section 26), isolated from the request-serving path

19.1 Zone Transition Rules

- Public → Application Zone: only through the Load Balancer, with authentication (Section 18) enforced before any request reaches Business Logic.
- Application → Restricted Zone: only through Repository components (Appendix-J Section 6) — no external caller ever issues a direct query against the Database Layer.
- Application → Integration Zone: only through the Event Publisher/ChannelAdapter pattern (Appendix-J Sections 6, 13), never a direct, synchronous call to an external vendor.

20. Encryption Strategy (Logical Only)

Reuses Appendix-F's Encryption NFR series (NFR-ENC-001/002) and Network Security series (NFR-NET-001/002) — described here at a logical level, with no specific cipher suite, key-management product, or vendor named.

Encryption Scope	Logical Requirement	Source
Data in Transit	All traffic encrypted end-to-end, Client Device to Application Layer and Application Layer to Database/Storage/Integration tiers	NFR-NET-001/002
Data at Rest — Sensitive/PII Fields	Field-level encryption for attributes flagged Sensitive/PII in Appendix-G/H (e.g., Student.medical_info, Employee.salary_structure_json, User.password_hash)	NFR-ENC-001
Data at Rest — File Storage	Encryption applied to documents linked to Sensitive/PII-classified entities (Section 10)	NFR-ENC-001
Backup Data	Backup artifacts (Section 16) encrypted and stored in a separate failure domain from primary infrastructure	NFR-BKP-001 (implied), NFR-ENC-001
Key Management	Logically required but mechanism/provider Client / Infrastructure Team Decision Required	N/A — Physical Design decision

21. Logging & Monitoring Architecture

Reuses Appendix-J Sections 10/21 (Logging Flow, Audit Trail), Appendix-F's Logging/Monitoring NFR series, and Appendix-L Sections 17-18, combined here as a single section per this document's structure.

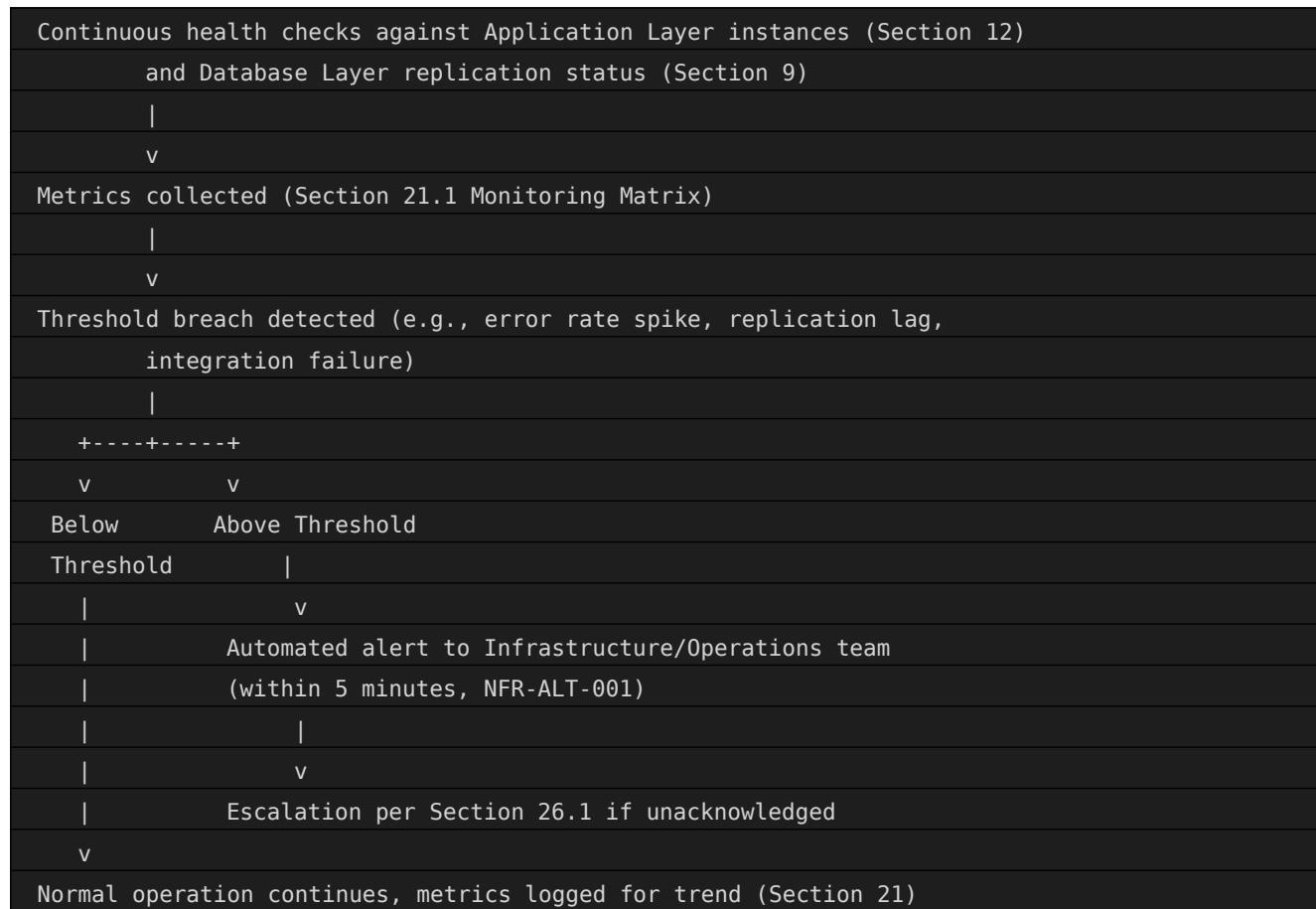
- Application-level logs (every module's Service layer, Appendix-J Section 10) and infrastructure-level logs (Load Balancer, Application Layer, Database Layer) are aggregated centrally rather than remaining siloed per tier.
- The AuditLog data path (Appendix-H ENT-SYS-004) is logically distinct from general infrastructure/application logging — audit records follow the retention and immutability rules in Appendix-H Section 15, while infrastructure logs follow a separate, typically shorter retention policy (Client / Infrastructure Team Decision Required).
- System Health Monitoring provides real-time visibility into uptime, response time, and error rate across the Application and Database Layers (NFR-MON-001).
- Third-Party Integration Monitoring is independent per external integration (NFR-MON-002) — a Payment Gateway outage is detected separately from an SMS Gateway outage, even though both route through the Integration Zone (Section 19).

21.1 Monitoring Matrix

Metric Category	Representative Metrics	Governing NFR
Availability	Uptime percentage, health-check success rate	NFR-AVAIL-001
Performance	Response time (p50/p95/p99), request throughput	NFR-PERF-001/011/012
Error Rate	Failed-request percentage, exception count by category	NFR-REL-001
Database	Query latency, connection-pool utilization, replication lag	NFR-DBP-001
Capacity	Concurrent-user count, CPU/memory utilization per tier	NFR-SCAL-001
Integration Health	Per-vendor success/failure rate and latency (Section 19)	NFR-MON-002
Security	Failed-login rate, account-lockout events, unusual-access-pattern flags	NFR-AUTHN-001, NFR-PWD-003
Backup	Backup job success/failure, restore-test outcome (Section 16)	NFR-BKP-001/005

22. Alerting & Health Checks

Reuses Appendix-F NFR-ALT-001 and Appendix-L Section 17, elaborated here as its own section with the required Monitoring & Alert Flow diagram.



- Health checks are the mechanism that allows the Load Balancer (Section 12) to route traffic away from an unhealthy Application Layer instance automatically, directly supporting the High Availability strategy (Section 13).
- Alerting infrastructure resides in the Management Zone (Section 19), isolated from the request-serving path so a monitoring-system issue cannot itself become a source of downtime.
- Alert thresholds, notification channels, and specific escalation timing remain Client / Infrastructure Team Decision Required.

23. Performance & Capacity Planning

Reuses Appendix-F's Performance NFR series and Appendix-L Sections 15.1-16.

Operation	Target (per Appendix-F)
Page Load Time	≤3 seconds (95th percentile, NFR-PERF-001)
API Response Time	≤500ms (95th percentile, NFR-PERF-011)
Database Query Response	≤200ms (95th percentile, indexed queries, NFR-PERF-012)
Fee Payment Processing	≤5 seconds gateway-confirmation-to-response (NFR-PERF-007)

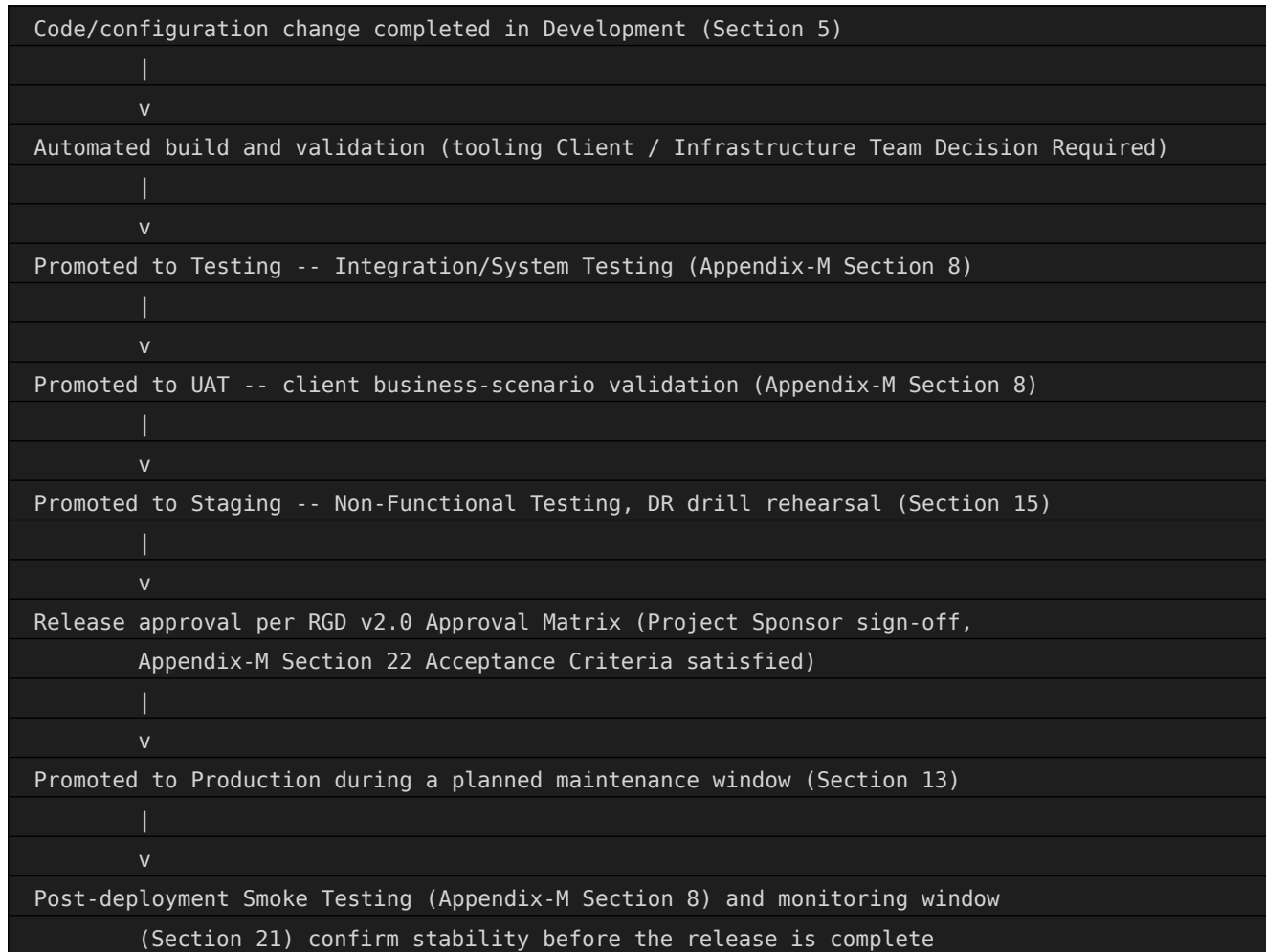
23.1 Growth Projection Reference

Reused directly from Appendix-H Section 17.7 — infrastructure sizing should plan against these figures, not initial go-live volume alone.

Entity	Approximate Annual Volume
Student (steady-state)	~20,000 active records
AttendanceRecord	10 million+ records/year
Invoice/Payment	500,000+ transactions/year
AuditLog	15 million+ records/year (estimated)

24. Release Deployment Flow

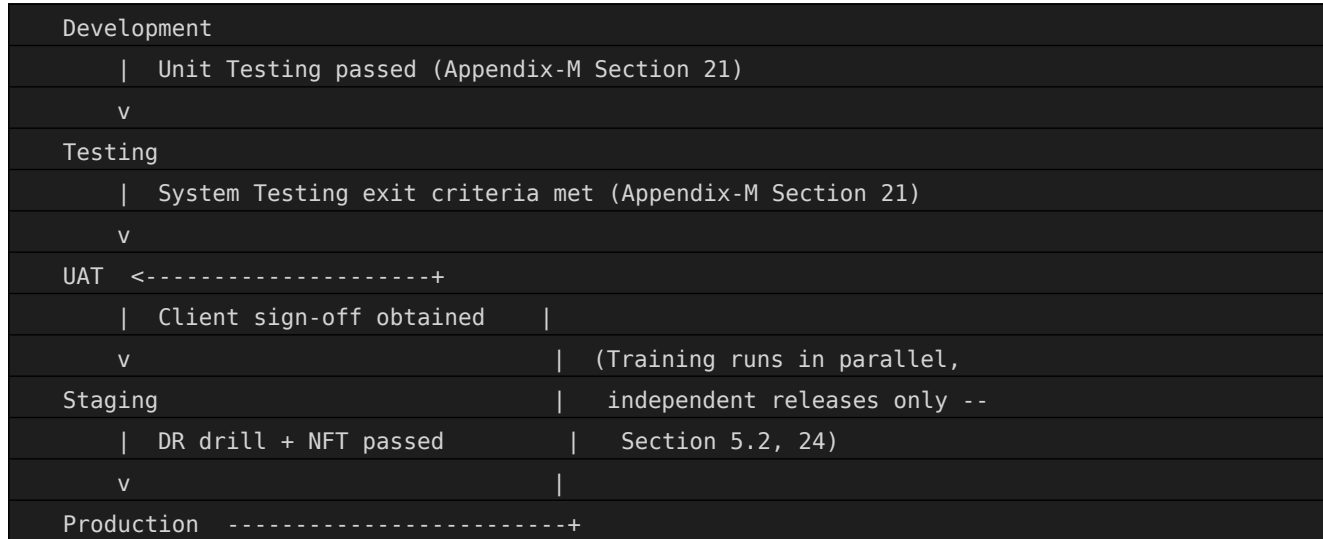
Reuses Appendix-L Section 19 (Deployment Lifecycle) and Section 20 (Release Strategy), presented here as the required Deployment Pipeline diagram.



- Training is not part of this promotion chain (Section 5.2) — it receives periodic, independent releases of stable, already-Production-validated functionality for rehearsal purposes.
- Release technique (blue-green, canary, rolling update, or maintenance-window cutover) remains Client / Infrastructure Team Decision Required — any is compatible with the stateless Application Layer design (Section 8).

25. Environment Promotion Process

Elaborates Section 5.2 with the required Environment Promotion Flow diagram.



- Each promotion step requires the corresponding Exit Criteria from Appendix-M Section 21 to be satisfied — this document does not redefine those criteria, only maps them onto the environment sequence.
- A failed promotion attempt returns the release candidate to Development for correction, per Appendix-M Section 16 Defect Life Cycle, rather than being force-promoted.

26. Operational Responsibilities

Reuses Appendix-L Section 22, extended here with explicit reference to the Test Strategy roles already defined in Appendix-M.

Responsibility	Owner (Client / Infrastructure Team Decision Required for named individuals/teams)
Infrastructure provisioning and change management	Infrastructure/Operations team
Database administration (Section 9)	Database Administrator role
Monitoring & Alerting response (Sections 21-22)	Infrastructure/Operations team, with Application-level escalation to IT Admin (Appendix-I Section 8)
Backup verification (Section 16)	Infrastructure/Operations team, per the monthly restore-test cadence
Disaster Recovery execution (Section 15)	Infrastructure/Operations team, coordinating with IT Admin for application-layer reconnection
Test execution across environments (Section 5)	QA Team, per Appendix-M Sections 6, 8
Release approval (Section 24)	Project Sponsor, per RGD v2.0 Approval Matrix
Training environment content/scheduling (Section 5)	IT Admin, coordinating with module owners (Appendix-G Data Ownership Matrix)

26.1 Escalation Path

A detected issue first alerts the directly responsible owner (Section 26 table); if unacknowledged within a configured window (Client / Infrastructure Team Decision Required), escalation proceeds to IT Admin, and for Production-impacting incidents, to the Project Sponsor per RGD v2.0's Approval Matrix for any decision requiring business-level authorization.

27. Deployment Validation Checklist

Check	Result
All 30 required sections present	PASS — Sections 1 through 30 as specified
All 6 environments detailed (Section 5)	PASS — Development, Testing, UAT, Training, Staging, Production
All 6 required ASCII diagrams present	PASS — Logical Infrastructure Architecture, Environment Promotion Flow, Deployment Pipeline, High Availability Architecture, Backup & DR Flow, Monitoring & Alert Flow
Minimum 10 required tables present	PASS — Environment Comparison, Infrastructure Components, Application Layer Responsibilities, Database Responsibilities, Storage Classification, Backup Schedule, Recovery Objectives, Monitoring Matrix, Operational Responsibility Matrix, Deployment Validation Checklist (this table)
No source code, SQL, APIs, or JSON/XML included	PASS
No Terraform, Kubernetes manifests, Docker Compose, shell commands, or configuration files included	PASS
No cloud vendor or product-specific technology named	PASS — every such decision marked Client / Infrastructure Team Decision Required
References to RGD and Appendix A–M included	PASS — Section 3
No new business processes, modules, or infrastructure invented	PASS — every reference traces to Appendix-F/H/I/J/K/L/M by ID

28. Infrastructure Risks & Assumptions

28.1 Risks

Risk	Source	Infrastructure Implication
Hosting provider not yet selected	RGD v2.0 Section 26; Appendix-L Section 23	Every component in Section 6.1 is designed as provider-agnostic to absorb this uncertainty
External integration vendors unselected	Appendix-I Section 11; Appendix-K Section 17	Integration Zone (Section 19) is isolated so vendor changes do not affect core infrastructure
High-volume table growth	Appendix-H Section 17; Appendix-L Section 23	Partition strategy (Sections 9, 14) must be operational from go-live
Six-environment model increases operational overhead versus Appendix-L's four-environment model	Section 5 (this document)	Additional environment (Training, UAT split from Staging) requires additional Section 26 operational capacity
RTO/RPO targets unconfirmed	Appendix-F NFR-BKP-003/004	Backup (Section 16) and DR (Section 15) infrastructure sizing cannot be finalized until confirmed

28.2 Assumptions

- Cloud-hosted deployment, per RGD v2.0 Business Assumptions (Section 22) — specific provider Client / Infrastructure Team Decision Required.
- A primary/replica relational database topology (Section 9) is assumed sufficient for the Scalability targets (Section 14) without requiring a distributed/sharded database architecture at initial go-live.
- The six environments (Section 5) are assumed sufficient for the project's delivery and operational lifecycle; no additional environment is assumed unless separately confirmed.
- All numeric/policy values already marked Client / Infrastructure Team Decision Required across Appendix-C v1.1, F, G, H, I, J, K, L, and M remain open at this infrastructure-design stage and are not resolved here.

29. Traceability Matrix

Appendix-N Section	Source Appendix
4. Deployment Objectives	Appendix-F NFR-AVAIL/REL/BKP/SCAL/PERF series
5. Environment Strategy	Appendix-L Section 4; Appendix-M Section 11 (extended with UAT/Training split)
6-8. Infrastructure, Topology, Application Layer	Appendix-I Section 7; Appendix-J Sections 5-6; Appendix-L Sections 5-6
9. Database Layer	Appendix-H Sections 12, 17; Appendix-L Section 10
10. File Storage Strategy	Appendix-I Section 12; Appendix-L Section 11
11-12. Network Segmentation, Load Balancing	Appendix-F Network Security series; Appendix-L Sections 7, 6.3
13-14. High Availability, Scalability	Appendix-F NFR-AVAIL/SCAL series; Appendix-L Sections 14-15
15-17. Disaster Recovery, Backup, Business Continuity	Appendix-F NFR-DR/BKP/BC series; Appendix-L Sections 12-13
18-20. Identity & Access, Security Zones, Encryption	Appendix-K Sections 4-5; Appendix-F Security series; Appendix-L Sections 8-9
21-22. Logging & Monitoring, Alerting	Appendix-J Sections 10/21; Appendix-F Monitoring/Logging series; Appendix-L Sections 17-18
23. Performance & Capacity Planning	Appendix-F Performance NFR series; Appendix-H Section 17.7
24-25. Release Flow, Environment Promotion	Appendix-L Sections 19-20; Appendix-M Section 8
26. Operational Responsibilities	Appendix-L Section 22; Appendix-M Section 18.1

Every section of this Appendix-N traces to at least one of RGD v2.0, Appendix-F, Appendix-H, Appendix-I, Appendix-J, Appendix-K, Appendix-L, or Appendix-M. No new business process, module, database object, API, or infrastructure element was introduced in producing this document.

30. Final Review

Metric	Value
Environments Covered	6 of 6 (Development, Testing, UAT, Training, Staging, Production)
Infrastructure Components Detailed	9 (Section 6.1)
Security Zones Defined	5 (Public, Application, Restricted, Integration, Management)
Required ASCII Diagrams	6 of 6 (Logical Infrastructure Architecture, Environment Promotion Flow, Deployment Pipeline, High Availability Architecture, Backup & DR Flow, Monitoring & Alert Flow)
Required Tables	10+ (Sections 5, 6.1, 8, 9, 10, 16, 16.1, 21.1, 26, 27)
Traceability to RGD/Appendix-F/H/I/J/K/L/M	Complete — Section 29
No Fabricated Scope	Confirmed — every module, entity, rule, API, and infrastructure element reused by reference
Deployment & Infrastructure Architecture Quality Score	90/100
Approval Status	Conditionally Ready — logical infrastructure design is structurally complete across all 6 environments and 30 sections; approval is gated on the consolidated Client / Infrastructure Team Decision Required items already tracked across Appendix-C v1.1, F, G, H, I, J, K, L, and M, plus the environment-specific decisions in Sections 5, 18.1, and 26 (Training environment scheduling, hosting provider, and operational team structure)

Deductions from a perfect score reflect the volume of hosting/vendor and operational decisions that remain legitimately open pending client/infrastructure-team input, not any structural incompleteness in this document's own 30 sections, 6 diagrams, or 10+ tables. This document formally closes Gap-14 and awaits review before any subsequent appendix is considered.